News Article Topic Classification

Arianna Giulia Biancone, Elena Nicolotti
Politecnico di Torino
Student id: s354965, s358639
s354965@studenti.polito.it, s358639@studenti.polito.it

***Abstract*—In this report, we address the problem of *news article topic classification*. The proposed approach relies on a hybrid feature representation that combines textual information, encoded through TF–IDF features, with categorical variables and context-related numerical information. Linear classification models are trained on the resulting representation and evaluated using the macro F1-score. The experimental results show that the proposed approach outperforms simple baseline solutions and demonstrates good generalization capability.**

I. Problem Overview

The project consists in predicting the topic associated with a news article. We consider the *News Articles Dataset*, which contains a total of 99,997 articles, together with their associated metadata. The dataset is divided into two parts:

- a *development set*, containing 79,997 articles along with their correct labels;
- an *evaluation set*, containing 20,000 articles for which the topic must be predicted.

The development set is used to build and validate a classification model, which is then employed to predict the topic of each article in the evaluation set.
Each article is described by a set of six attributes, excluding the target variable. These include both unstructured textual features, such as the article body and title, and structured metadata. An overview of the available features and their cardinality is reported in Table I.

TABLE I
Dataset features

Name	Type	Cardinality
Id	Numerical	79997
Source	Nominal	1359
Title	Text	75783
Article	Text	74393
Page_rank	Numerical	4
Timestamp	Nominal	52183

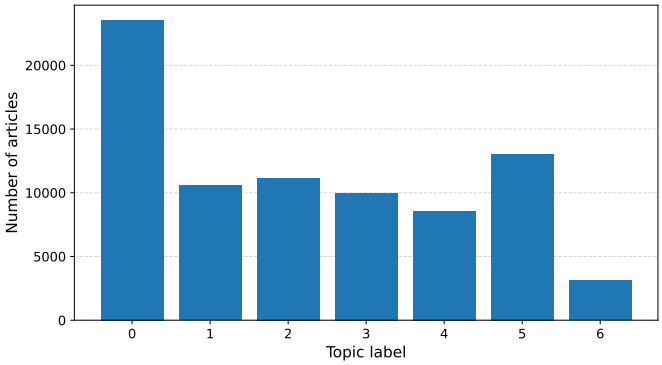


Fig. 1. Distribution of topic labels in the development set

The target variable corresponds to the topic associated with each article. Its distribution in the development set is shown in Figure 1. As can be observed, the dataset exhibits a significant class imbalance, which represents a relevant challenge for the classification task.

In addition, the dataset presents several data quality issues. The textual content of many articles contains HTML tags, which do not convey semantic information and must therefore be explicitly handled to avoid introducing noise into the learning process. Moreover, duplicated entries are present in the dataset, and in some cases identical articles are associated with different topic labels, leading to label ambiguity. Finally, the *Timestamp* attribute presents inconsistencies, as several values correspond to invalid dates (i.e., "0000-00-00 00:00:00").

These characteristics motivate the preprocessing strategy adopted in this work.

II. Proposed Approach

A. Preprocessing

1) Data cleaning: the dataset was first examined for integrity issues, leading to the removal of records with null or empty values in the *Article* column. To maintain a clean development set, we first removed duplicate records with identical *Title* and *Article* content sharing the same label, as they do not provide additional information. Furthermore, instances with identical textual content and associated with different labels were entirely removed to avoid introducing label noise.

2) Text refinement: a dedicated cleaning function, `clean_html`, was applied to a textual corpus obtained by concatenating the *Source*, *Title*, and *Article* fields into a single text representation. This combined text constitutes the textual input used for training the model. In our approach, the *Source* information was processed twice: first as part of the textual corpus to contribute to the overall textual representation, and subsequently as a standalone categorical feature. This choice reflects the observation that news publishers often exhibit strong topical biases, as they tend to focus on specific subjects, making the source a potentially informative signal for topic classification.

The cleaning process of the text is composed of several steps. First, the text was normalized to lowercase and HTML

entities were unescaped using the `html` library. Then, regular expressions were employed to remove HTML tags and non-alphabetical characters, retaining only characters in the range `a--z`. Single-character words were removed to reduce vocabulary sparsity and eliminate terms with limited power.

Finally, samples whose cleaned textual content was shorter than 20 characters were discarded, as such inputs typically lack sufficient semantic information to support reliable classification.

3) Temporal and numerical processing: the *Timestamp* attribute contained several invalid entries coded as "0000-00-00 00:00:00". Missing or invalid temporal information was explicitly encoded through a new binary feature `date_missing`. For valid timestamps, the attributes *Year* and *Day_of_week* were extracted and treated as numerical features, as they may capture both long-term temporal trends and recurring weekly patterns in news topics (e.g., sports events often occurring on specific days of the week).

In addition to temporal information, a numerical feature representing the length of each article was included, as document length may convey useful contextual suggestions. To reduce imbalance and limit the influence of extreme values, the word count was log-transformed, yielding the feature

$$\texttt{Num_word_log} = \log(1 + \texttt{Num_words}).$$

4) Feature Extraction: to transform the cleaned data into a numerical format, we utilized a `ColumnTransformer` to apply a hybrid strategy:

- Textual vectorization: the textual content was encoded using a TF–IDF representation, in order to emphasize terms that are frequent within a document but relatively rare across the corpus. This improves discriminative power for topic classification. To capture both local lexical patterns and short contextual expressions, multiple n-gram configurations were considered. Additional mechanisms were employed to control vocabulary size and sparsity, such as document frequency filtering and sublinear term-frequency scaling. Moreover, standard lexical normalization steps were applied, such as English stopword removal and Unicode accent normalization, to reduce noise and improve vocabulary consistency. The selection process of the TF–IDF hyperparameters is discussed in Section C
- Numerical scaling: the structured variables *Page_rank*, *Year*,*Day_of_week* and *Num_word_log*, were first imputed using a `SimpleImputer` with the mean strategy and subsequently standardized through a `StandardScaler`.
- Categorical encoding: the *Source* column was imputed using a `SimpleImputer` with the most frequent strategy and then transformed via One-Hot Encoding. The encoder was configured to ignore unknown categories during evaluation to ensure pipeline stability.

B. Model selection

The following classification algorithms have been selected for the considered task:

- **Logistic Regression**: we adopted this model due to its effectiveness in handling the high-dimensional sparsity of TF–IDF features [1]. In this task, documents are represented in a high-dimensional feature space, for which logistic regression provides a simple and effective probabilistic modeling of class membership. Regularization and class weighting were employed to control overfitting and to account for class imbalance.
- **SVM**: we selected the LinearSVC implementation to explicitly exploit a margin-based learning criterion, which is well suited for high-dimensional sparse text representations [2], [3]. By focusing on maximizing the separation between classes rather than modeling class probabilities, this approach provides a complementary perspective to logistic regression. In line with the previous model, we applied class weighting to mitigate the effects of class imbalance.

Hyperparameters for the two models were selected through a grid-based search using a K-Fold cross-validation strategy, as explained in the following section.

C. Hyperparameters tuning

The hyperparameters considered in the tuning process can be grouped into two main categories:

- hyperparameters related to the TF–IDF representation adopted in the preprocessing stage;
- hyperparameters associated with the Logistic Regression and Linear SVM classifiers.

Hyperparameter tuning was performed using a grid-based search combined with a 3-fold cross-validation strategy. The tuning process was applied jointly to the preprocessing stage and the classification models by embedding the TF–IDF representation and the classifier within a single pipeline. This approach ensured that hyperparameters related to feature extraction and model learning were optimized consistently, while preventing information leakage across validation folds.

For each configuration, performance was evaluated using the macro-averaged F1 score. Table II summarizes the hyperparameter configurations explored during the tuning phase.

TABLE II
HYPERPARAMETERS CONSIDERED IN THE TUNING PHASE

Model	Parameter	Values
Preprocessing (TF–IDF)	ngram_range min_df max_df	{(1,2), (1,3)} {2, 5} {0.30, 0.50}
Logistic Regression	C class_weight	{ 1, 4, 5, 10, 20} {balanced}
Linear SVM	C class_weight	{0.1, 0.5, 1, 5, 10} {balanced}

III. Results

The tuning process led to the same optimal preprocessing configuration for both models, namely $\{$`ngram_range` $= (1, 3)$, `min_df` $= 2$, `max_df` $= 0.30\}$, indicating a consistent preference for richer contextual representations across classifiers. In addition, the parameter `class_weight` was fixed to `balanced` in order to mitigate the effects of class imbalance.

Figures 2 and 3 report the mean cross-validation macro F1-score as a function of the regularization parameter C, considering only the subset of configurations that match the selected TF–IDF setup.

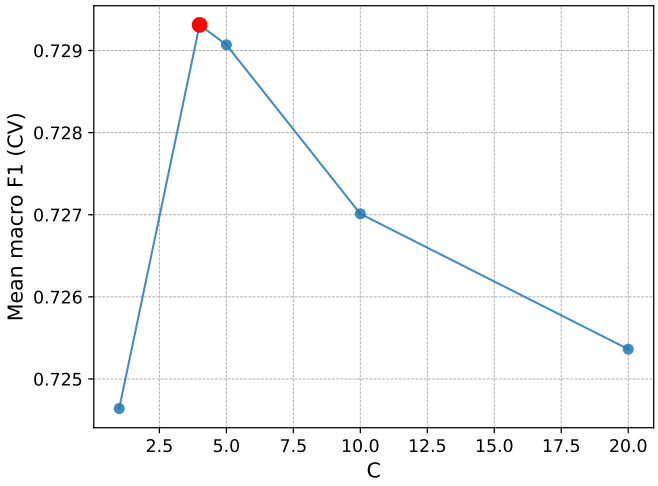


Fig. 2. Logistic Regression: Mean macro F1-score (CV) as a function of C.

As a result of the tuning process, the selected regularization strengths were $C = 4$ for Logistic Regression, achieving a macro F1-score of 0.729, and $C = 0.5$ for the Linear SVM, with a macro F1-score of 0.730.

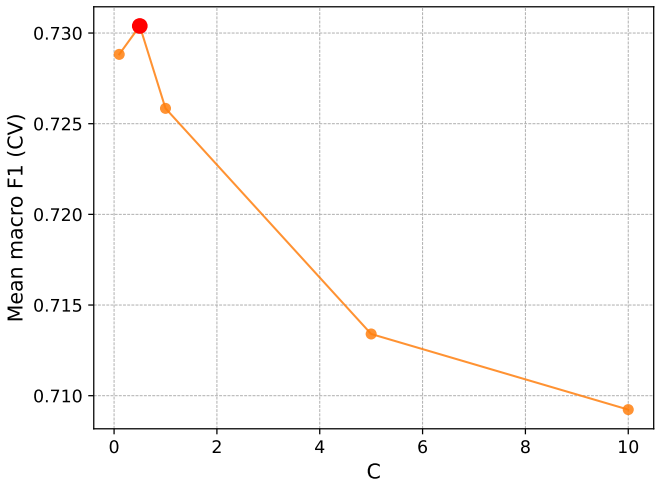


Fig. 3. Linear SVC: Mean macro F1-score (CV) as a function of C.

After training on the entire development set, both Logistic Regression and Linear SVM achieved a public evaluation macro F1-score of 0.732 on the evaluation set. The close agreement between the validation and public evaluation performance indicates good generalization capability for both classifiers, and should indicate no overfitting. Moreover, the very similar performance levels achieved by Logistic Regression and Linear SVM suggest that no clear advantage emerges from one model over the other for the considered task.

To contextualize our results, we evaluated two naive baseline configurations:

1) **Most-frequent baseline**: implemented via a dummy classifier, this approach consistently predicts the majority class. It achieves a macro F1-score of approximately 0.42, establishing a lower bound for our task.
2) **Source and Title baseline**: this model combines the *Source*, treated as a categorical feature, with the article *Title*, encoded using the same TF–IDF representation adopted in the full model. As expected, access to the title and the source already provides substantial semantic information, yielding a macro F1-score of 0.684 with Logistic Regression and 0.652 with Linear SVM. Nevertheless, the performance gap with respect to our proposed approach confirms the added value of leveraging the full article content and contextual features in improving class discrimination.

Furthermore, if we compare our best score of 0.732 with the other submissions on the leaderboard, it appears to be well positioned overall. Excluding a small number of top-performing submissions, which reach scores around 0.745, the remaining top entries differ only at the third decimal place, indicating that the performance gap among most solutions is marginal.

IV. Discussion

The experimental results indicate that, within the class of linear models, performance is largely driven by the quality of the feature representation and preprocessing strategy. These observations motivate several possible directions for future improvements:

- Lemmatization: in order to reduce lexical variability of words and therefore provide a more sematincally consistent tf-idf representation.
- Dimensionality reduction techniques: to find a more compact representation of the data while preserving most of its variability. In particular, PCA-based method, such as Truncated SVD, could be explored.
- Deep Learning: transitioning to non-linear architectures, such as Neural Networks, could help capture complex relationships between features that linear models may overlook.

Overall, the results obtained are very promising when compared to the baseline set for the project.

REFERENCES

[1] K. Shah, H. Patel, D. Sanghvi, *et al.*, "A comparative analysis of logistic regression, random forest and knn models for the text classification," *Augmented Human Research*, vol. 5, no. 12, 2020.

[2] T. Joachims, "Text categorization with support vector machines," in *Proceedings of the International Conference on Machine Learning (ICML)*, 1998.

[3] Z. Liu, X. Lv, K. Liu, and S. Shi, "Study on svm compared with the other text classification methods," in *Proceedings of the Second International Workshop on Education Technology and Computer Science*, IEEE, 2010.